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EG3013S Chip User Manual

High-power MOSFET and IGBT gate driver chips

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Version Change Log

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Table of contents

1. Features..... 4
2. Description..... 4
3. Application areas..... 4
4. Pins..... 4
4.1. Pin Definitions..... 4
4.2. Pin Description..... 4
5. Structural diagram.....5
6. Typical Application Circuits..... 5
7. Electrical characteristics..... 7
7.1 Limiting Parameters..... 7
7.2 Typical Parameters..... 8
7.3 Switching Time Characteristics and Dead Time Waveform 9
8. Application Design..... 10
8.1 Vcc terminal power supply voltage..... 10
8.2 Input logic signal requirements and output driver characteristics..... 10
8.3 Bootstrap Circuit..... 11
9. Package dimensions..... 12
9.1 SOP8 package dimensions..... 12

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EG3013S Chip Datasheet **V1.0**

1. Features

• High-end floating bootstrap power supply design, withstand voltage up to 80V

• Built-in dead-time control circuit • Built-in

latch-up function, completely preventing simultaneous conduction of upper and lower transistor

outputs • Utilizes a half-bridge Darlington transistor output structure with 1A high-current gate drive capability •

Dedicated for N-channel MOSFET and IGBT gate drive in brushless motors • HIN input channel

active high level, controlling the high-side HO output • LIN input channel active low level,

controlling the low-side LO output • Few external components • Low quiescent current: 4.5mA •

Package: SOP-8

2. Description

The EG3013S is a high-performance, cost-effective dedicated chip for driving the gates of high-power MOSFETs and IGBTs, and it integrates logic signal input processing.

The circuit includes a dead-time control circuit, a latching circuit, a level shifting circuit, a pulse filtering circuit, and an output drive circuit, specifically designed for brushless motor control.

The drive circuit in the device.

The EG3013S can operate at up to 80V on the high-end, with a wide Vcc power supply range of 4.5V to 30V and a low quiescent power consumption of only 4.5mA. This chip features...

It has a latching function to prevent the output power transistors from conducting simultaneously. The input channel LIN has a built-in 5V high-potential pull-up and HIN has a built-in 15K pull-down.

The resistor keeps the upper and lower power MOSFETs off when the input is floating. The output structure adopts a half-bridge Darlington transistor structure and is packaged in an SOP8 package.

3. Application Areas

Electric motorcycle controller **Electric**

bicycle controller **80V** step-down switching

power supply

ÿ Variable frequency water pump

controller ÿ Brushless motor driver ÿ

High voltage Class-D power amplifier

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4. Pins

4.1. Pin Definitions

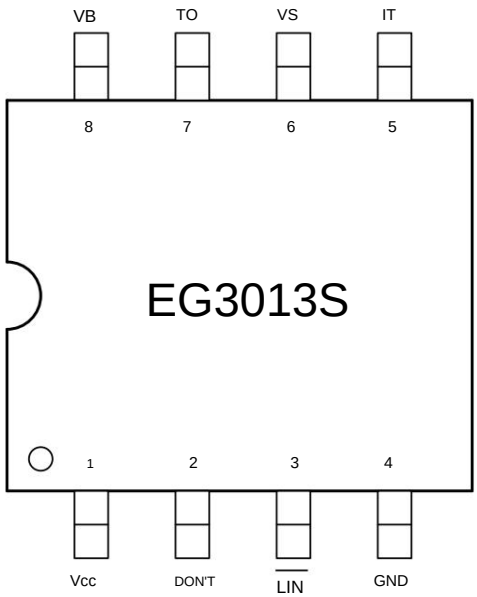


Figure 4-1. EG3013S Pin Definitions

4.2. Pin Description

Pin number	Pin name	I/O	describe
1	Vcc	Power	The recommended typical operating voltage for the chip's power input is 4.5V-15V. An external power supply is required. A 0.1uF bypass capacitor at high frequencies can reduce high-frequency noise at the chip's input.
2	DON'T		The logic input control signal is active high, controlling the on/off state of the high-side power MOSFET. Stop. When the LIN pin is high, the HIN function is as follows (see Section 8.2 for details). "0" means the power MOSFET is turned off. "1" indicates that the power MOSFET is enabled.
3	$\overline{\text{LIN}}$		The logic input control signal is active low, controlling the on/off state of the low-side power MOSFET. When the HIN pin is low, the LIN function is as follows (see Section 8.2 for details). "0" indicates that the power MOSFET is enabled. "1" means turn off the power MOSFET.
4	GND		GND is the ground terminal of the chip.
5	IT		O output controls the on/off state of the low-side MOSFET power transistor.
6	VS		O High-end suspended ground end
7	TO		O output controls the on/off state of the high-side MOSFET power transistor.
8	VB		Power High-end Floating Power Supply

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5. Structural Diagram

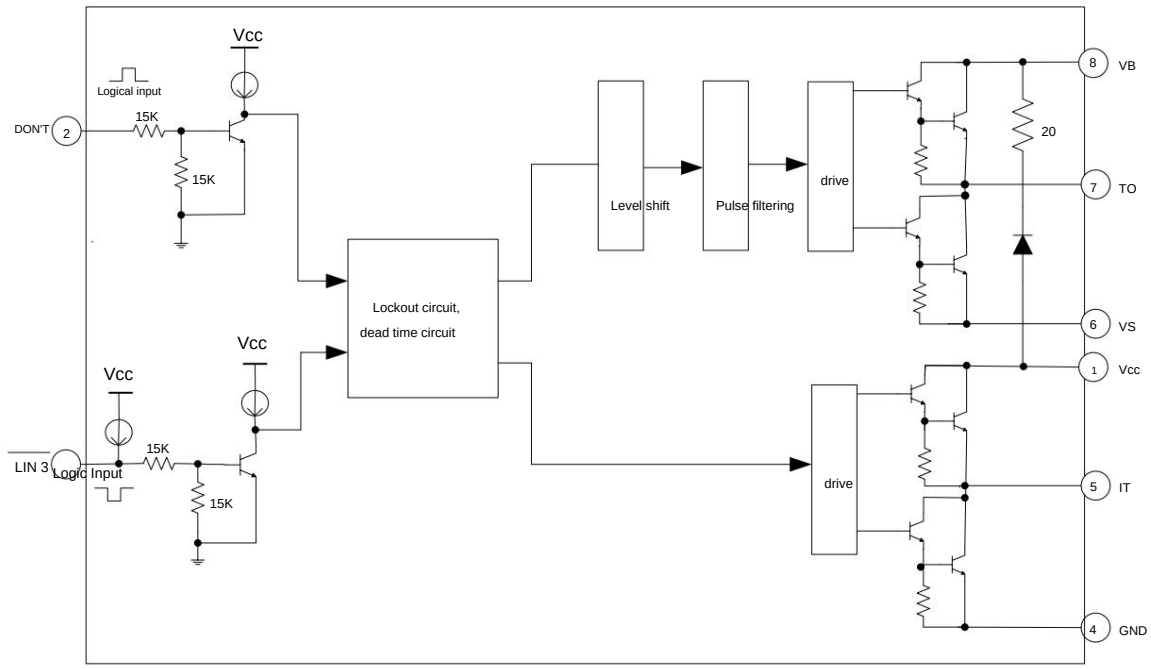


Figure 5-1. Block diagram of EG3013S

6. Typical Application Circuits

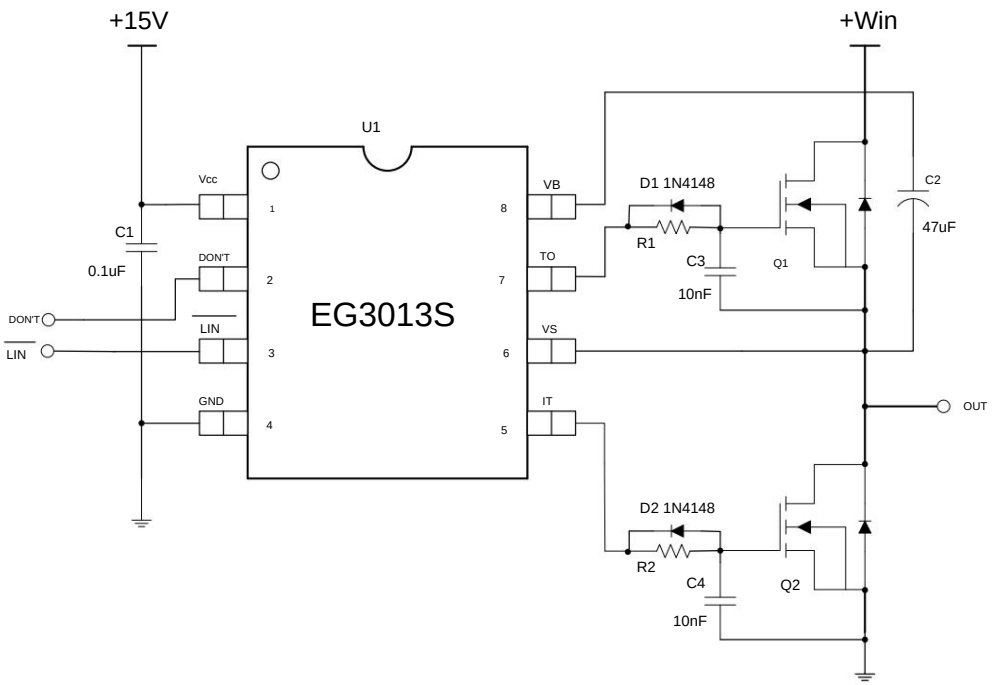


Figure 6-1. Typical application circuit diagram of EG3013S - medium and low power half-bridge drive application

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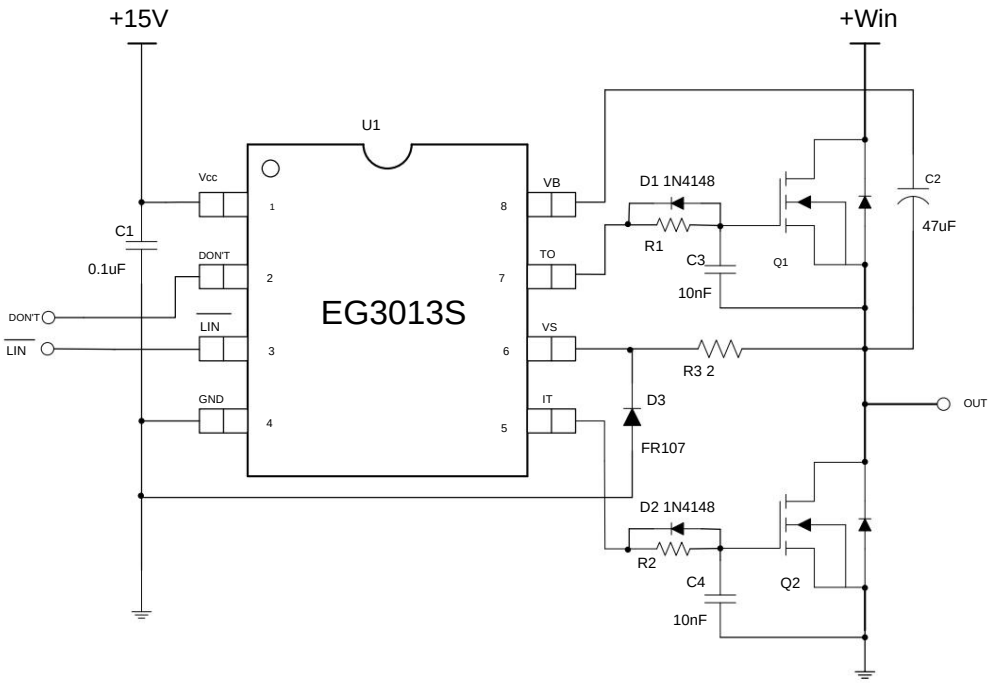


Figure 6-2. Typical application circuit diagram of EG3013S – High-power motor application

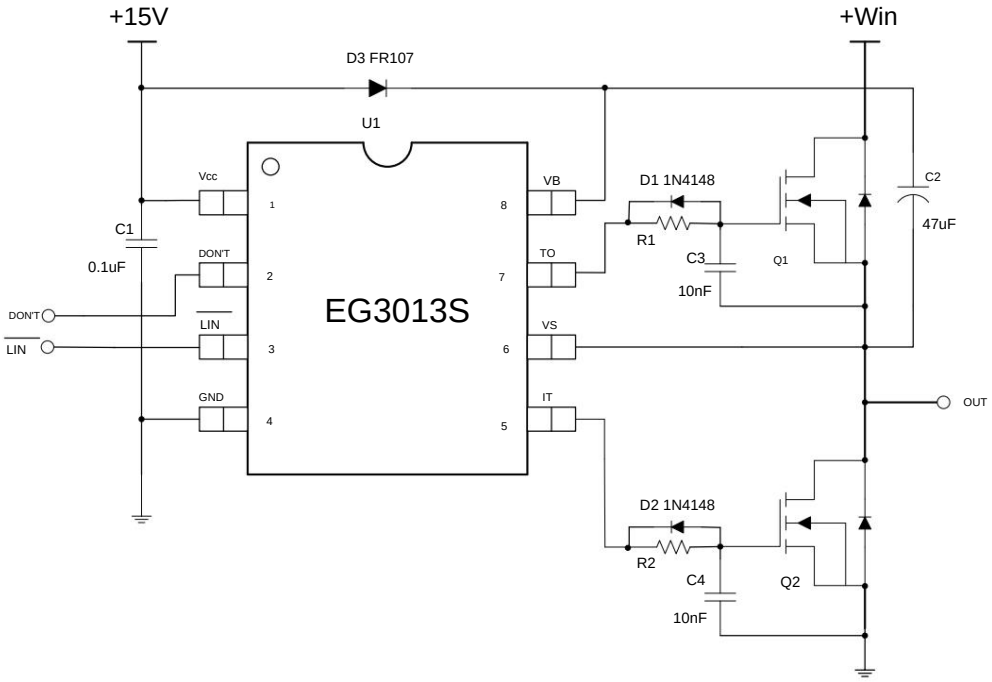


Figure 6-3. Typical application circuit diagram of EG3013S - application of external bootstrap diode.

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7. Electrical characteristics

7.1 Limiting Parameters

Unless otherwise specified, under the condition of TA=25ÿ

Symbol Parameter Name	Bootstrap High-end VB Power	Test conditions minimum and maximum	units		
VB	Supply		-0.3	90	In
VS	High-end suspended ground		-0.7	80	In
TO	High-end output		-0.3	90	In
IT	Low-end output		-0.3	35	In
VCC	High-		-0.3	35	In
DON'T	channel power logic signal output Input level		-0.3	35	In
LIN	LIN low-channel logic signal input level		-0.3	35	In
FACING	Ambient temperature		-45	125	ÿ
Tstr	Storage temperature		-55	150	ÿ
TL	welding temperature	T=10S		300	ÿ

Note: Exceeding the listed limits may cause permanent damage to the chip's internal components. Prolonged operation under extreme conditions may affect the chip's reliability.

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7.2 Typical Parameters

Unless otherwise specified, under the conditions of TA=25ÿ, Vcc=15V, and load capacitance CL=10nF.

Parameter Name Symbol Power Supply		Test conditions minimum typical maximum unit				
	Vcc		4.5	15	30	In
Static current	Icc input left floating, Vcc=15V			4.5	6	m.a.
input logic signal high Low	Vin(H) All input control signals 2.5			5.0		In
potential input logic signal High	Vin(L) -0.3 for all input control signals.			0	1.0	In
potential input logic signal Low level current	Yes(H)	Vin=5V		300	400	a
input logic signal Level current	Yes(L)	Vin=0V		0		a
Low-side output (LO) switching time characteristics						
Enable delay	Ton	See Figure 7-1		500	700	nS
Guan Yinshi	Toff	See Figure 7-1		50	100	nS
Ascent Time	Tr	See Figure 7-1		400	600	nS
Descent time	Tf	See Figure 7-1		200	300	nS
High-end output HO switching time characteristics						
Enable delay	Ton	See Figure 7-2		300	500	nS
Guan Yinshi	Toff	See Figure 7-2		400	600	nS
Ascent Time	Tr	See Figure 7-2		400	600	nS
Descent time	Tf	See Figure 7-2		200	300	nS
Dead time characteristics						
Dead Time	DT	See Figure 7-3. No-load capacitor CL=0	80	130	400	nS
Maximum drive capability of IO output						
IO output current IO+		Vo=0V,Vin=VIH PWÿ10uS	0.6	0.8		A
IO output sink current IO-		Vo=15V,VIN=VIL PWÿ10uS	0.8	1		A

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7.3 Switching time characteristics and dead time waveform diagram

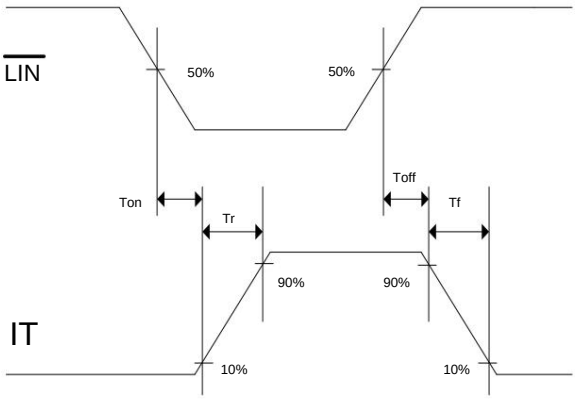


Figure 7-1. Low-end output LO switching time waveform diagram

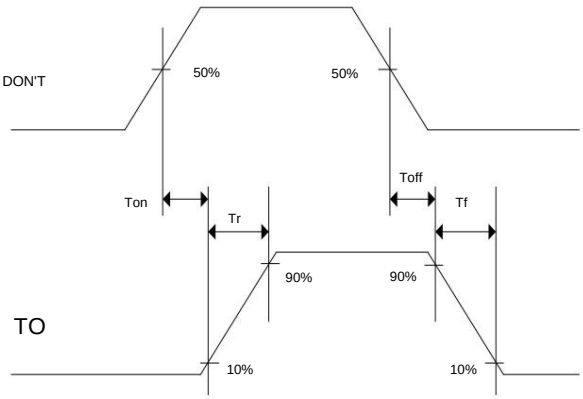


Figure 7-2. High-side output HO switching time waveform diagram

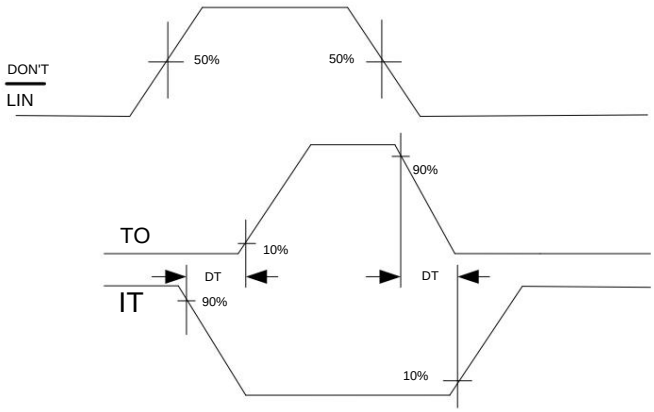


Figure 7-3. Dead time waveform diagram

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8. Application Design

8.1 Vcc terminal power supply voltage

To ensure sufficient drive voltage for the N-channel power MOSFET, a typical operating voltage of 4.5V-15V is recommended for the power supply Vcc. (Internal logic...)

The power supply for the logic circuit and the analog level conversion circuit share the same Vcc power supply, and the internal logic ground and analog ground are also connected together.

8.2 Input Logic Signal Requirements and Output Driver Characteristics

The EG3013S's main functions include logic signal input processing, dead-time control, level conversion, a floating bootstrap power supply structure, and upper/lower bridge totem-pole outputs. The high-level threshold for the logic signal input is above 2.5V, and the low-level threshold is below 1.0V. It requires low output current for the logic signals, allowing the MCU's output logic signals to be directly connected to the EG3013S's input channels.

The totem-pole output drivers for both the high-end upper arm and low-end lower arm have a maximum sink current of 1A and a maximum output current of 0.8A. The high-end upper arm channel can withstand 80V. The conduction delay between the input logic signal and the output control signal is small: the low-end output turn-on conduction delay is 500ns and the turn-off conduction delay is 100ns; the high-end output turn-on conduction delay is 300ns and the turn-off conduction delay is 400ns. The low-end output turn-on rise time is 400ns and the turn-off fall time is 200ns; the high-end output turn-on rise time is 400ns and the turn-off fall time is 200ns.

The logic function diagram for the input and output signals is shown in Figure 8-2:

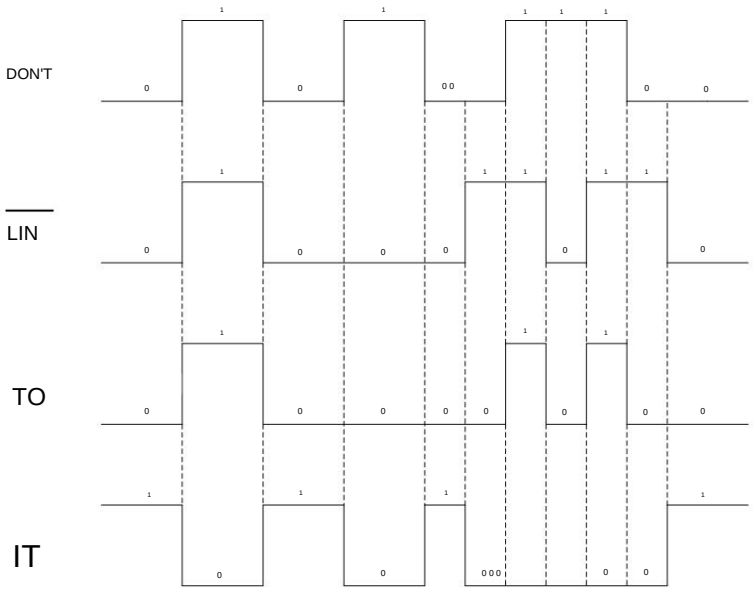


Figure 8-2. Logic function diagram of input and output signals

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Input and output signal logic truth tables:

enter		Output	
Input and output logic			
HIN (pin 2)	LIN (pin 3) HO (pin 7)		LO (pin 5)
0	**0**	0	**1**
0	1	0	0
1	0	0	0
1	**1**	**1**	0

From the truth table, it can be seen that when the input logic signals HIN and LIN are not simultaneously "0" and not simultaneously "1", the driver controls the output HO.

When LO is simultaneously "0", both the upper and lower power transistors are turned off; when the input logic signals HIN and LIN are simultaneously "0", the driver control output HO is...

When the upper transistor is "0", it is off; when LO is "1", the lower transistor is on. When both input logic signals HIN and LIN are "1", the driver control output HO is "1".

When the upper transistor is on and the LO is "0", the lower transistor is off; the internal logic processor prevents the upper and lower power transistors at the controller output from being on simultaneously, and has a mutual interlocking function.

able.

8.3 Bootstrap Circuit

The EG3013S employs a bootstrap floating drive power supply structure, greatly simplifying drive power supply design. It can complete high-end power supply operation using only one power supply voltage, VCC.

The driving of two power switching devices, an N-channel MOSFET and a low-side N-channel MOSFET, greatly simplifies practical applications. The EG3013S can enable...

The bootstrap voltage boost function is automatically completed using an internal bootstrap diode or an external bootstrap diode as shown in Figure 8-3 and a bootstrap capacitor. It is assumed that the lower transistor is on.

During the on/off period of the upper transistor, the bootstrap capacitor C has been charged to a sufficient voltage (Vc=VCC). When HO outputs a high level, the upper transistor is turned on and the lower transistor is turned off.

The voltage across the bootstrap capacitor VC will act as an equivalent voltage source to power the internal drivers VB and VS, thus driving the high-side N-channel MOSFET.

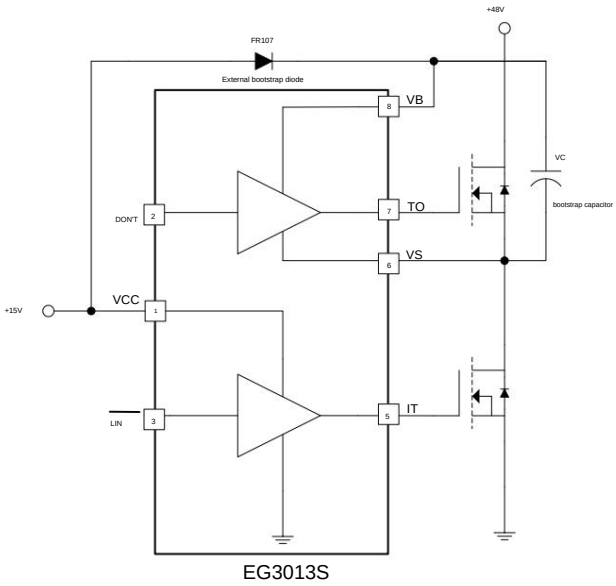


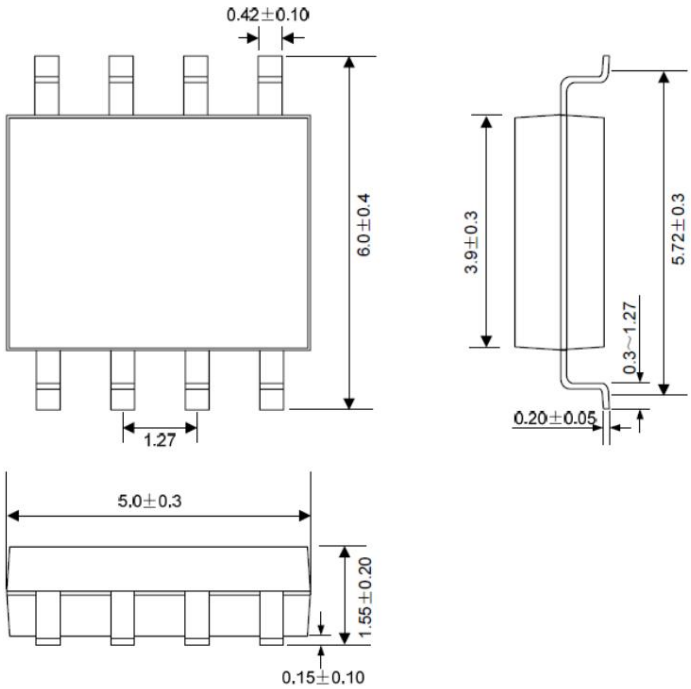
Figure 8-3. EG3013S bootstrap circuit structure

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9. Package size

9.1 SOP8 package dimensions



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